

Also, $(8 + 0 + 8 + 7) - (3 + 6 + 3) = (23 - 12) = 11$.

So, the given number is divisible by 11.

- 215.** $(2 + 4 + 9 + 8 + 4) = 27$, $(2 + 6 + 7 + 8 + 4) = 27$ and $(2 + 8 + 5 + 8 + 4) = 27$.

So, each one is divisible by 3 and 9 both.

Also, 84 is divisible by 4. So, each one is divisible by 4.

Hence, each given number is divisible by 3, 9 and 4.

So, the statements 1, 2 and 3 are all correct.

- 216.** We know that 872 is divisible by 8. Hence 923872 is divisible by 8.

- 217.** Here $(5 + 9 + x + 7) - (4 + 3 + 8) = 6 + x$. So, we must have $x = 5$.

- 218.** Take $m = 15$ and $n = 20$. Then, each one of m and n is divisible by 5. But, $(m + n)$ is not divisible by 10.

Hence, $(m + n)$ is divisible by 10 is not true.

- 219.** An integer is divisible by 16, if the number formed by last 4 digits is divisible by 16.

- 220.** Clearly, 639 is not divisible by 7.

Consider 2079. Sum of its digits $= (2 + 0 + 7 + 9) = 18$.

So, it is divisible by both 3 and 9.

Also, $(79 - 2) = 77$, which is divisible by 7.

So, 2079 is divisible by 7.

Also, $(9 + 0) - (7 + 2) = 0$. So, 2079 is divisible by 11.

Hence, 2079 is divisible by each one of 3, 7, 9 and 11.

- 221.** Let the given number be $476xy0$. Then

$$(0 + x + 7) - (y + 6 + 4) = 0 \Rightarrow x - y - 3 = 0 \Rightarrow x - y = 3.$$

Also, $(4 + 7 + 6 + x + y + 0) = (17 + x + y)$ must be divisible by 3.

Since $x \neq 0$, $y \neq 0$, so $x + y \neq 1$.

$\therefore x + y = 4$ or 7 or 10 etc.

$(x + y = 4 \text{ and } x - y = 3) \Rightarrow x = 7/2$, which is not admissible.

$(x + y = 7 \text{ and } x - y = 3) \Rightarrow x = 5$ and $y = 2$.

- 222.** Sum of the digits in respective numbers is:

9, 12, 18, 9, 21, 12, 18, 21, 15 and 24.

Out of these 12, 21, 12, 21, 15, 24 are divisible by 3 but not by 9.

So, the number of required numbers is 6.

- 223.** Let the unit's place be x and the thousand's place be y . Then, $357y25x$ is divisible by 5 only when $x = 0$ or $x = 5$. Also, this number is divisible by 3 only when sum of its digits is divisible by 3.

So, $(22 + x + y)$ must be divisible by 3.

$\therefore x + y = 2$

Taking $x = 0$, we get $y = 2$.

So, the unit place = 0 and thousand's place = 2.

- 224.** Clearly, $(7 + 8) - (9 + 6) = 0$. So, 6897 is divisible by 11.

Also, $\frac{6897}{19} = 363$. So, 6897 is divisible by 19.

Hence, 6897 is divisible by both 11 and 19.

- 225.** We have $24 = 3 \times 8$, where 3 and 8 are co-primes.

Clearly, 718 is not divisible by 8. So, 35718 is not divisible by 8.

810 is not divisible by 8. So, 63810 is not divisible by 8.

804 is not divisible by 8. So, 537804 is not divisible by 8.

736 is divisible by 8. So, 3125736 is divisible by 8.

Also, sum of its digits $= (3 + 1 + 2 + 5 + 7 + 3 + 6) = 27$, which is divisible by 3.

So, 3125736 is divisible by 3 also.

Hence, it is divisible by 24.

- 226.** Sum of the digits of the given number $= (7 \times 3) + (14 \times 1) = (21 + 14) = 35$, which is not divisible by 3.

So the given number is not divisible by 3.

Also, $(1 + 3 + 1 + 1 + 3 + 1 + 1 + 3 + 1 + 1 + 3) - (1 + 1 + 3 + 1 + 1 + 3 + 1 + 1 + 3 + 1) = (19 - 16) = 3$, which is neither 0 nor divisible by 11.

So, the given number is not divisible by 11 also.

Hence, it is divisible by neither 3 nor 11.

- 227.** $(325 - 325) = 0$, which is divisible by 7.

So, the given number is divisible by 7.

$(5 + 3 + 2) - (2 + 5 + 3) = 0$. So, the given number is divisible by 11.

And, $\frac{325325}{13} = 25025$. So, 325325 is divisible by 13.

Hence, it is divisible by all 7, 11 and 13.

- 228.** We first divide the number into groups of 3 digits from the right $\rightarrow 3 \text{ } 0X0 \text{ } 103$

Difference of sum of numbers at odd and even places $= (103 + 3) - 0X0 = 106 - 0X0$, which must be divisible by 13.

$106 - 0X0$ is divisible by 13 only for $X = 8$.

- 229.** We know that 11 and 13 are co-prime.

So, a number divisible by both 11 and 13 will be divisible by (11×13) .

- 230.** I. We have $(195 - 195) = 0$

$\therefore 195195$ is divisible by 7.

II. We have $(181 - 181) = 0$

$\therefore 181181$ is divisible by 7.

III. We have $(120 - 120) = 0$

$\therefore 120120$ is divisible by 7.

IV. We have $(891 - 891) = 0$

$\therefore 891891$ is divisible by 7.

Hence, all are divisible by 7.

- 231.** Since $653xy$ is divisible by 80, we must have $y = 0$.

Now, $653x0$ must be divisible by both 5 and 16.

Clearly, it is divisible by 5 for all values of x .

Now, the number $53x0$ must be divisible by 16.

The least value of x is clearly 6. So, $x + y = 6 + 0 = 6$.

- 232.** We know that $33 = 11 \times 3$, where 11 and 3 are co-primes.

So, the given number must be divisible by both 11 and 3.

Since $5ABB7A$ is divisible by 11, we have

$(A + B + A) - (7 + B + 5) = (2A - 12)$ is either 0 or 11.

$$\Rightarrow 2A - 12 = 0 \text{ or } 2A - 12 = 11 \Rightarrow A = 6 \left[\because A \neq \frac{23}{2} \right]$$

So, the number becomes $56BB76$, which is divisible by 3.

$\therefore (5 + 6 + B + B + 7 + 6) = (24 + 2B)$ must be divisible by 3.

$$\therefore 2B = 6 \Rightarrow B = 3 \quad \left[\because B \neq 0 \text{ and } B \neq \frac{3}{2} \right]$$

Hence, $(A + B) = (6 + 3) = 9$.

- 233.** We have, $99 = (11 \times 9)$, where 11 and 9 are co-primes.

Consider the number 114345.

Clearly, $(5 + 3 + 1) - (4 + 4 + 1) = 0$. So, 114345 is divisible by 11.

Also, sum of its digits $= (1 + 1 + 4 + 3 + 4 + 5) = 18$, which is divisible by 9.

$\therefore 114345$ is divisible by 9.

Hence, it is divisible by (11×9) , i.e. 99.

- 234.** Let the unit's digit be x and ten's digit be y . Then, the number is $3422213yx$.

Also, $99 = (11 \times 9)$, where 11 and 9 are co-primes.

Since the given number is divisible by 9, it follows that

$(3 + 4 + 2 + 2 + 2 + 1 + 3 + y + x) = (17 + y + x)$ must be divisible by 9.

So, $y + x = 1$ or $y + x = 10$.